

Generative AI education: Building a community of practice at Mayo Clinic

生成式 AI 教育：在 Mayo Clinic 建立一個實踐社群

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出處：Schultz, K. E., Van Erp, N. L., Minter, C. J., & Nelson, K. B. (2026). Generative AI education: Building a community of practice at Mayo Clinic. *Innovations in Education and Teaching International*, 63(3), 882–887.

DOI：10.1080/14703297.2025.2507299

公信力等級：Q1 期刊（SSCI 收錄，Education 類別），影響指數約 4.9，SJR 約 0.81。但本篇作者明言其為「opinion piece（觀點文）」，非實證研究；宜當「實務佐證、概念來源、研究動機背書」，不宜當「方法論主幹」。

研讀範圍：本版為全文收錄（摘要、第 1 至 5 節、Table 1、Figure 1、2、作者簡介與參考文獻均含）。

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Abstract | 摘要

Due to the rapid evolution of generative AI, we recommend a minimalist community of practice model to promote continuous social learning in an environment that fosters discussion, inquiry and collaboration while minimising administrative burden. To illustrate the relative ease with which such a model can be established and sustained, this paper describes an interdisciplinary collaborative effort by educators from multiple departments at Mayo Clinic. While we have encountered challenges, we are heartened by the engagement of our community as well as the unexpected benefits we enjoy as the founding members and facilitators of the community.

中文譯文 面對生成式 AI 的快速演進，我們主張採用一種極簡的實踐社群(community of practice)模式，在鼓勵討論、提問與協作的環境中促進持續的社會性學習，同時把行政負擔降到最低。為了說明這種模式建立與維繫起來其實相對容易，本文描述了 Mayo Clinic 多個部門的教育工作者所進行的一場跨領域協作。一路上我們確實遇到一些挑戰，但社群的投入，加上我們身為創始成員與引導者所享受到的意外收穫，都令我們深受鼓舞。

KEYWORDS: Generative AI; health professions education; higher education; learning communities; communities of practice

中文譯文 關鍵詞：生成式 AI；健康專業教育；高等教育；學習社群；實踐社群

1. Introduction/Background | 1. 前言／背景

Every so often something new comes along that changes what is possible. In November 2022, OpenAI released ChatGPT and sparked the proliferation of a new technology that raised many questions for educators:

中文譯文 每隔一段時間，就會出現某種改變「什麼是可能」的新事物。2022 年 11 月，OpenAI 推出 ChatGPT，點燃了一波新技術的擴散，也為教育工作者帶來許多問題：

- What are the risks of generative AI?

中文譯文 生成式 AI 有哪些風險？

- How do you keep up with and apply these emerging tools?

中文譯文 你要怎麼跟上並實際運用這些新興工具？

- How do you grow confidence with tools that are changing constantly?

► 中文譯文 面對不斷在變的工具,你要怎麼一步步建立信心?

This opinion piece explains why we believe the answer lies in the community. It describes how our experiences illustrate that the community of practice (CoP) model is a viable option for any group in higher education that wants to take an active role in growing their confidence and competency with generative AI.

► 中文譯文 這篇觀點文要說明的是,為什麼我們相信答案就在「社群」裡。我們透過自身經驗來呈現:對任何想主動提升自己在生成式 AI 上的信心與能力的高等教育群體而言,實踐社群(CoP)模式都是一個可行的選項。

In October 2023, we launched a virtual Generative AI in Education CoP at Mayo Clinic to bring together healthcare educators to learn about generative AI by asking questions, sharing knowledge and discussing practical use cases. A CoP is a group with a common interest that meets periodically to share ideas and engage in a rich dialogue to problem-solve and advance the collective understanding of a topic (de Carvalho-Filho et al., 2020). Although the CoP concept has evolved over time (Li et al., 2009), its core idea of social learning in an organisation to promote knowledge-sharing is as relevant as ever. It can be challenging to learn about tools that do not come with user manuals and change constantly, particularly if they come with ethical and policy implications. A CoP is a space that brings to life evidence-based and applied learning (Brooks et al., 2024). Our CoP realises those ideals: it is practical, meaningful and community-driven, while grounded in both theory and practice. What is unique about our CoP, however, is how easy it has been to establish and maintain, contrasting other CoP models (Brooks et al., 2024).

► 中文譯文 2023 年 10 月,我們在 Mayo Clinic 推出了一個線上的「教育中的生成式 AI」實踐社群,把醫療照護領域的教育工作者聚在一起,透過提問、分享知識與討論實際使用情境來認識生成式 AI。所謂實踐社群,是一群擁有共同興趣的人,定期聚會交流想法、進行豐富的對話,藉此一起解決問題並深化對某個主題的集體理解(de Carvalho-Filho et al., 2020)。儘管實踐社群這個概念隨時間不斷演變(Li et al., 2009),它的核心精神,也就是在組織內透過社會性學習來促進知識分享,至今依然非常切合需要。要去學一套沒有使用手冊、又持續在變的工具本來就不容易,尤其當這些工具還牽涉倫理與政策層面時更是如此。實踐社群正是一個讓「以證據為本」與「應用導向」的學習真正活起來的空間(Brooks et al., 2024)。我們的社群實現了這些理想:它務實、有意義、由社群驅動,同時又扎根於理論與實務。不過,我們這個社群最特別的地方在於,相較於其他實踐社群模式,它建立與維繫起來竟然如此容易(Brooks et al., 2024)。

Over the past year, we learned about establishing and sustaining a CoP responsive to the needs of our participants while supporting the digital transformation goals of our institution. We also learned that there are unexpected benefits for us as facilitators. We are excited to share our lessons learned, including how our CoP is structured, what works well, our challenges and the overall power of learning in a community.

► 中文譯文 過去這一年,我們學到了如何建立並維繫一個既能回應參與者需求、又能支援機構數位轉型目標的實踐社群。我們也發現,身為引導者的我們其實得到了一些意想不到的好處。我們很樂意分享這些心得,包括我們的社群是怎麼組織起來的、哪些做法行得通、遇到了什麼挑戰,以及在社群中一起學習所帶來的整體力量。

2. Building the community | 2. 建立社群

Our facilitation team brings unique perspectives from our diverse roles at Mayo Clinic, united by our collective goal to grow understanding of generative AI in education. Our monthly planning process is dynamic, adapting to ever-evolving generative AI tools, topics and guidance. Our roles emerged organically, dividing the responsibilities of preparing and promoting upcoming sessions. All four of us co-facilitate discussions, which minimises disruptions when one of us cannot attend a session.

► 中文譯文 我們的引導團隊來自 Mayo Clinic 不同的職務角色,各自帶來獨特的觀點,而把我們凝聚在一起的,是「提升大家對教育中生成式 AI 的理解」這個共同目標。我們每個月的籌備流程很有彈性,會隨著不斷演進的生成式 AI 工具、主題與相關指引調整。我們的分工是自然而然形成的,把籌備與宣傳下一場活動的責任分攤開來。四個人都會共同引導討論,這樣一來,即使有人臨時無法出席,活動也不太會受到影響。

We intentionally keep our CoP sessions informal to create a space where attendees feel comfortable asking questions, sharing insights and building connections. Each month, our CoP sessions became more sophisticated and nuanced as

participants' knowledge of generative AI grew. The progression highlights how the community not only learns together but also shapes the content itself, an essential principle of CoPs (de Carvalho-Filho et al., 2020). This progression also serves as a responsive way to address the sense of never being up-to-speed when tools change constantly and are impossible to master completely (Craft, 2024).

中文譯文 我們刻意讓社群活動維持輕鬆非正式的氛圍,營造一個讓參與者能自在提問、分享心得、建立連結的空間。隨著參與者對生成式 AI 的認識日漸加深,我們每個月的活動內容也變得愈來愈深入細緻。這樣的演進凸顯了一件事:社群不只是一起學習,更會親手形塑學習的內容本身,而這正是實踐社群的一項核心原則 (de Carvalho-Filho et al., 2020)。這種逐步推進的方式,同時也是一帖良方,用來化解那種「工具一直在變、永遠不可能完全精通,因此總覺得自己跟不上」的焦慮(Craft, 2024)。

3. Successes and challenges | 3. 成果與挑戰

As we complete the first year of our CoP, we are reflecting on our successes and challenges thus far. We are proud of the range of topics and formats we offered for our sessions (Table 1).

中文譯文 在社群滿一年的此刻,我們回頭檢視這段期間的成果與挑戰。我們對自己在活動中所提供的多元主題與形式感到自豪(見 Table 1)。

To keep the content fresh and engaging, we alternate topic themes from month-to-month. Our attendance grew steadily throughout the year (Figure 1), largely through word-of-mouth. Every session included new members, indicating an extension in our reach. We also see the impacts of our CoP demonstrated within our sessions when our members present use cases inspired by their learnings from previous sessions.

中文譯文 為了讓內容保持新鮮、引人投入,我們每個月輪替不同的主題類型。整年下來,出席人數穩定成長(見 Figure 1),主要是靠口耳相傳。每一場都有新成員加入,顯示我們觸及的範圍正在擴大。當成員把前幾場學到的東西轉化成實際使用案例、在活動中分享出來時,我們也親眼看見了社群帶來的影響。

Table 1. Community of practice session topic themes from the first year. 表 1. 實踐社群第一年的活動主題類型。

Topics 主題類型	# of Sessions 場次數
'How to' generative AI tool demonstrations 「如何操作」生成式 AI 工具示範	3
Use case presentations 使用案例分享	3
Generative AI in education podcast discussions 教育中的生成式 AI 播客討論	3
Opportunities and challenges to implementing generative AI discussions 導入生成式 AI 的機會與挑戰討論	2
Ethics of generative AI 生成式 AI 的倫理	2

中文譯文 第一年我們共辦了 12 場活動。不過其中一場涵蓋多個主題,因此本表加總為 13。工具示範的例子包括:為醫學教育生成圖像、用 ChatGPT 摘要播客逐字稿並擬出討論題,以及用 Perplexity AI 來尋找並彙整資源。使用案例分享則展示了 Mayo Clinic 的教育同仁如何在工作中運用生成式 AI。

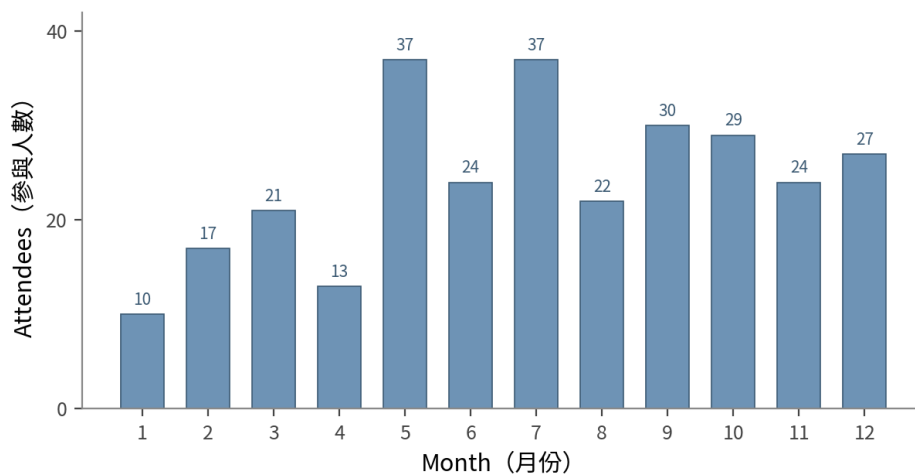


Figure 1. Number of attendees over 1 year. 圖 1. 一年間的參與人數。

(本圖依原文 Figure 1 長條高度判斷後乾淨重繪,走勢與原圖一致;確切數值以原文為準。)

We frequently seek attendee feedback to provide members with a sense of shared ownership of their learning (Rainer & Matthews, 2002) and to help plan sessions. With so many potential topics as generative AI quickly evolves, feedback from members helps us prioritise the most salient and urgent topics. Our members are eager to explore applications of generative AI within education at Mayo Clinic. We often hear from colleagues that they are excited by AI but overwhelmed; they are unsure where to begin or how to engage, particularly given organisational constraints and concerns about data privacy. Our CoP reduces some of these barriers by centring data security and our organisation's regulations in our conversations. It is one thing to learn about a trendy new AI tool, but it is another to see its direct use in health professions education. By learning from peers about how generative AI is being applied at our institution, rather than simply learning what generative AI can theoretically do, participants are better prepared to integrate it into their work.

► **中文譯文** 我們經常向參與者徵詢回饋,一方面讓成員對自己的學習產生「共同擁有」的感覺(Rainer & Matthews, 2002),一方面也用來協助規劃後續活動。由於生成式 AI 演進飛快、可談的主題實在太多,成員的回饋能幫我們把最切身、最迫切的主題排出優先順序。我們的成員都很想探索生成式 AI 在 Mayo Clinic 教育場域中的各種應用。我們也常聽同事說,他們對 AI 既興奮又不知所措,不確定該從哪裡開始、該怎麼參與,尤其在組織的種種限制與資料隱私的顧慮之下更是如此。我們的社群把資料安全與機構規範擺在對話的核心,藉此降低了其中一部分門檻。畢竟,認識一個正夯的新 AI 工具是一回事,親眼看到它怎麼直接用在健康專業教育上又是另一回事。比起單純去了解生成式 AI「理論上能做什麼」,透過向同儕學習它在我們機構裡「實際上怎麼被使用」,參與者更能做好準備,把它整合進自己的工作。

One early challenge we noted was attrition associated with breakout rooms. We wanted to leverage breakout rooms because they encourage deeper conversation among learners (Read et al., 2022). However, we noticed several attendees left the session when breakout rooms began. This is not unique to our CoP, as we noticed this in other virtual gatherings as well. We know breakout sessions can be anxiety-inducing and that participants may not be able to engage in breakout discussions if they are in a space where they cannot use their microphone or camera. Therefore, we planned full-group sessions so community members could participate by speaking up, using the chat tool or by observing. This flexibility is key, allowing for engaging conversations while providing a low-stress environment for attendees.

► **中文譯文** 我們早期注意到的一個挑戰,是分組討論室(breakout rooms)造成的流失。我們本來想善用分組討論室,因為它能促進學習者之間更深入的對話(Read et al., 2022)。然而我們發現,一旦開始分組,就有好幾位參與者直接離開了。這並不是我們社群獨有的現象,我們在其他線上聚會也觀察到同樣的情況。我們明白,分組討論可能讓人焦慮,而且如果參與者所處的環境不方便開麥克風或鏡頭,他們也很難真正投入分組對話。因此,我們改採全體一起進行的形式,讓社群成員可以選擇開口發言、用聊天室文字參與,或者單純在旁觀看。這種彈性是關鍵,既能保有熱絡的對話,又能为參與者營造一個低壓力的環境。

Lastly, a known challenge in building a CoP is the resources required to successfully launch and maintain the community, with some organisations using dedicated staff/departments to support CoPs (Brooks et al., 2024). However, we found that with deliberate planning, a CoP can thrive with minimal administrative burden on its facilitators by distributing the workload and leveraging existing tools within the institution. Moreover, we found unexpected benefits in our facilitator

planning meetings; regularly brainstorming together has improved our sense of belonging within our organisation. Building strong relationships and being known by colleagues are important to remote workers' sense of belonging (Dery & Hafermalz, 2016), and this experience of establishing and co-facilitating a CoP around a topic that is important to us increased our connection. Facilitating the CoP creates a feeling of doing our work 'together', even when working within our separate teams. Furthermore, it helps us build relationships with fellow educators who we may not have otherwise met, allowing for additional formal collaboration and informal connection outside of the CoP sessions.

中文譯文 最後,建立實踐社群還有一個眾所周知的挑戰,就是成功啟動與維繫社群所需投入的資源,有些組織甚至會動用專責人力或部門來支援(Brooks et al., 2024)。然而我們發現,只要事前刻意規劃,透過分攤工作量、善用機構內既有的工具,一個實踐社群其實可以在引導者行政負擔極低的情況下蓬勃運作。不僅如此,我們還在引導團隊的籌備會議中得到了意外的收穫:固定一起腦力激盪,讓我們對組織的歸屬感變得更強。對遠距工作者而言,建立穩固的關係、被同事認識,對歸屬感至關重要(Dery & Hafermalz, 2016),而這段一起圍繞著我們共同在乎的主題、共同建立並引導社群的經驗,確實拉近了彼此的連結。引導這個社群讓我們即使各自待在不同團隊,仍有一種「一起在做事」的感覺。更進一步,它還幫我們和一些原本可能沒機會認識的教育同行建立起關係,讓社群活動之外多了正式合作與非正式往來的可能。

4. Lessons learned | 4. 學到的心得

We attribute the success of our CoP to four factors that we believe can aid others interested in establishing a CoP:

中文譯文 我們把社群的成功歸結為四個因素,相信這些因素也能幫助其他有意建立實踐社群的人:

(1) Establish a small team of volunteer leaders – We need to be attuned to the developments in generative AI and potential implications and uses in education. With four facilitators, the administrative load for staying on top of trends in generative AI, planning and facilitating sessions is manageable and enjoyable (Figure 2).

中文譯文 (1)建立一支小型的志願領導團隊:我們必須對生成式 AI 的發展,以及它在教育上可能帶來的影響與用途保持敏銳。有四位引導者一起分攤,要跟上生成式 AI 趨勢、規劃並引導活動的行政負擔就變得可以負荷,甚至樂在其中(見 Figure 2)。

(2) Maintain an informal approach – Keeping our effort informal keeps it simple. Each month we meet once to plan the session, complete assigned tasks and then facilitate the session.

中文譯文 (2)維持非正式的運作方式:把整件事保持得輕鬆非正式,就能保持簡單。我們每個月碰一次面規劃活動、各自完成分配到的任務,然後把活動辦起來。

(3) Use existing tools – We use Microsoft Teams, email and a SharePoint site to communicate and organise. This allows us to leverage what is available without additional effort.

中文譯文 (3)善用既有工具:我們用 Microsoft Teams、電子郵件與一個 SharePoint 網站來溝通與整理事務。這讓我們不必另起爐灶,就能直接運用手邊現成的資源。

(4) Complement other efforts – Our CoP is one of the several efforts within our organisation to promote generative AI education. Working together with those efforts, while setting a clear intention and audience for our initiative, has amplified their impacts.

中文譯文 (4)與其他行動相輔相成:我們的社群只是組織內推動生成式 AI 教育的眾多行動之一。一方面與這些行動協力合作,一方面為我們自己的倡議設定清楚的目標與受眾,反而把彼此的影響力都放大了。

5. Future directions | 5. 未來方向

As we look to the future of our CoP at Mayo Clinic, we will build upon the foundation we established over the past year. We will be formally evaluating our CoP to directly measure the impact our sessions have had on our participants' work at Mayo Clinic using the Value-Creation Framework (Wenger-Trayner et al., 2019). Our journey thus far has shown us the power of community-driven learning, and we are committed to expanding our impact while staying true to our core principles of informally yet intentionally fostering knowledge exchange and connection.

中文譯文 展望 Mayo Clinic 這個社群的未來,我們會在過去一年打下的基礎上繼續往上蓋。我們將正式評估這個社群,運用價值創造框架(Value-Creation Framework)直接衡量我們的活動對參與者在 Mayo Clinic 工作上的實際影響(Wenger-Trayner et al., 2019)。一路走到現在,這段旅程讓我們親身體會到社群驅動式學習的力量。我們會持續擴大影響力,同時忠於我們的核心原則:以非正式卻有意圖的方式,促成知識的交流與人與人的連結。

Mayo Clinic Generative AI in Education CoP Facilitator's Minimalist Framework

Mayo Clinic 「教育中的生成式 AI」實踐社群引導者的極簡框架

Three Weeks Prior to Session: Facilitator Planning Meeting (one hour) | 活動前三週: 引導者籌備會議 (一小時)

- ☐ Debrief previous CoP session
檢討上一場社群活動
- ☐ Discuss plans for next session, including identifying general area of focus and format (e.g., AI tool demonstrations and guided practice sessions, podcast or journal discussion, use case presentations). Identify potential guest presenters based on topics selected, if needed
討論下一場的規劃, 包括確定大方向主題與形式 (例如 AI 工具示範與引導實作、播客或期刊討論、使用案例分享)。視需要, 依選定主題物色可能的客座講者
- ☐ Note additional ideas for future sessions as they emerge via a shared planning document (e.g., Microsoft OneNote)
隨時把未來活動的點子記到共用的規劃文件裡 (例如 Microsoft OneNote)

Two Weeks Prior to Session: Asynchronous Preparation | 活動前兩週: 非同步準備

- ☐ Write CoP description
撰寫社群活動說明
- ☐ Confirm guest presenters (as needed) and send calendar invites to presenters
視需要確認客座講者, 並寄出行事曆邀請給講者
- ☐ Discuss any remaining logistics via facilitator's group chat on Microsoft Teams
透過 Microsoft Teams 的引導者群組聊天室討論其餘并務

One Week Prior to Session: Promotion | 活動前一週: 宣傳推廣

- ☐ Post CoP session description on event intranet/SharePoint
把社群活動說明張貼到活動內部網站/SharePoint
- ☐ Email reminder with session description to all registered attendees
寄送含活動說明的提醒信給所有報名者
- ☐ Post reminder and description in AI Education Teams Channel
在 AI 教育的 Teams 頻道張貼提醒與說明

Immediately After Session: Wrap-Up | 活動結束後立即: 收尾整理

- ☐ Post recording on event intranet/SharePoint site, which allows anyone within Mayo Clinic to access past recordings, topics, and resources from our sessions
把錄影上傳到活動內部網站/SharePoint, 讓 Mayo Clinic 內部任何人都能取得歷次活動的錄影、主題與資源
- ☐ Record attendance information and note suggestions and chat topics from session to inform future session ideas
記錄出席資訊, 並記下活動中的建議與聊天室話題, 作為未來活動的點子來源
- ☐ Use facilitator chat to continue discussions of potential topics prior to next planning meeting
在下次籌備會議前, 用引導者聊天室持續討論可能的主題

Figure 2. Mayo Clinic generative AI in education community of practice facilitator's minimalist framework. 圖 2. Mayo Clinic 「教育中的生成式 AI」實踐社群引導者的極簡框架。

(Figure 2 原為純文字的籌備框架, 此處依原文位置以中英對照重建, 以便閱讀; 若需原圖截圖可另行提供原 PDF 檔。)

We encourage educators, faculty developers and leaders to consider how a minimalist CoP model may work for your organisation. As we continue to learn about generative AI, and as the technology evolves, the power of learning together can transform how we approach our work while promoting both a sense of community and innovation in higher education.

中文譯文 我們鼓勵教育工作者、師資培育者與領導者,認真想想一套極簡的實踐社群模式能如何在你的組織裡運作。隨著我們持續認識生成式 AI、隨著技術不斷演進,「一起學習」這股力量,將能改變我們面對工作的方式,同時在高等教育中孕育出社群感與創新。

Disclosure statement | 利益衝突聲明

No potential conflict of interest was reported by the author(s).

► 中文譯文 作者聲明無任何潛在利益衝突。

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